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Started	Saturday, 17 October 2026, 11:59 PM
Completed	Sunday, 18 October 2026, 12:04 AM
Duration	4 mins 51 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process

b. It allows for quicker identification of duplicates

c. It makes the analysis slower

d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **2**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Using the divide-and-conquer approach
- c. Being the most efficient sorting algorithm
- d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. All of the above
- c. Selection
- d. Duplicates

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. Each part is sorted independently
c. The parts are ignored
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **5**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By using the bubble-up method
c. By sorting data in a single pass
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking

b. Dynamic programming

c. Divide and conquer

d. Greedy method

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Bubble sort

c. Insertion sort

d. Heap sort

The correct answer is:
Insertion sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is less efficient than bubble sort
- b. It is based on the divide-and-conquer approach
- c. It does not require recursion
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **9**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Merging
- b. Sorting
- c. Complexity
- d. Searching

The correct answer is:
Complexity

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data in a single pass
c. Dividing the input into parts, solving each part, and combining the solutions
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging
b. Merging
c. Searching
d. Sorting

The correct answer is: Sorting

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It only works on small datasets
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Bubble Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. order()
- c. arrange()
- d. sorted()

The correct answer is: sort()

Question **15**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys